Results in Special Relativity, a Geometrical approach

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Abstract

The aim of these notes is to present the core of Special Relativity (SR), and how it is embedded in Minkowski Spacetime $\mathbb{M}^4$ and its Lorentzian metric g. The first section serves as a justification of the affine structure of the manifold $\mathbb{M}^4$ and the Poincarè Group as the class of transformations from one inertial reference frame to another. The proof of all the results presented here can be found in V. Moretti's Analytical Mechanics [1].

Postulates and justification of the geometrical approach

Special Relativity (SR), owes its birth to the inconsistency between Electromagnetism and Galilean relativity, as d'Alembert equations (of the form $\Box^2\psi = 0$) is not invariant under the Galilei Group. The historical justification of this inconsistency was that there is a preferred reference frame, called "Aether", in which d'Alembert equations hold. But since Michelson and Morley showed that the speed of light is isotropic, if such a reference frame existed, it would be at rest with respect to Earth at any time. Therefore the hypothesis of Aether was discarded. The structure of spacetime itself, considered as a 4-dimensional C^∞ manifold $\mathbb{M}^4$, needed to be modified.

Rulers and clocks We assume to deal with ideal rulers and clocks. The property of being ideal means that if they are at rest in some inertial reference frame, they have the same length and measure time in the same way, and this feature will persist if they happen to be at rest in some inertial reference frame again, independently of their individual worldlines. It's fundamental to note that this property doesn't imply that the *synchronization* of clocks, seen as an equivalent relation between clocks, is necessarily preserved if the worldlines of clocks are different.

Synchronization problem The synchronization procedure isn't unique, and the formulation of SR depends on it. More precisely, different ways of synchronizing distant clocks lead to (potentially) different theories. *Einstein's synchronization* works in the following way:

1. Clocks A and B are at rest in some inertial reference frame, at a given distance l;
2. A sends a signal of given speed v to B[1];
3. B is adjusted, taking into consideration that the signal took a finite amount of time to reach B.

We can now list SR postulates:

> SR1: there exist reference frames in which, given a defined synchronization procedure, the speed of light assumes the same value, and such value is constant in time, and doesn't depend on the direction or specific frame.

> SR2: the class of reference frames described in (SR1) is made of all (and only) inertial reference frames, where isolated material points move at constant velocity.

> SR3: all physical laws have the same form in every inertial reference frame.

Minkowski spacetime

Thanks to (SR1) and (SR2) it can be shown that, given two systems of Minkowskian coordinates (coordinates adapted to inertial reference frames), the transformation between the two has always the form:

$$x'^{\alpha} = b^{\alpha} + \Lambda^{\alpha}_{\beta} x^{\beta} \qquad \Lambda^T \eta \Lambda = \eta, \quad \alpha = 0,1,2,3 \tag{1}$$

where $\eta = \text{diag(-1,1,1,1)}$. Since the laws of transformation between Minkowskian coordinates are linear, through equations (1), where (b, Λ) form a group called **the Poincarè Group**, $\mathbb{M}^4$ can be equipped with the affine structure and indefinite product described by η, which can be seen as a global (and intrinsic) metric. This observation justifies the following approach: we start from $\mathbb{M}^4$, equipped with the structures mentioned earlier, and recover SR as a direct consequence of the geometry of spacetime.

Definition 1. The 4-dimensional C^∞ manifold $\mathbb{M}^4$ is called **Minkowski spacetime**, its points are called **events**, the metric g on its **vector space of translations**, which has signature (1,3), is called **Minkowskian metric**.

[1] in principle, one would need two clocks to measure v and that requires synchronization, but if the signal follows a closed path this problem is avoided, as only one clock is needed.

We now want to define a particular subset of applied vectors in p: $\mathcal{V}_p \subset T_p\mathbb{M}^4$.

Definition 2. $\forall p \in \mathbb{M}^4$, we can define the **lightcone** in p: $\mathcal{V}_p = \{(p,v) \in T_p\mathbb{M}^4 | v \in V^4, g(v,v) < 0\}$. The elements of $\overline{\mathcal{V}_p} \setminus \{0\}$ are called **causal vectors in p**, and can be divided in:

1. **null-like** if g(v,v)=0;
2. **time-like** if g(v,v)<0;
3. **space-like** if $v \notin \overline{\mathcal{V}_p} \setminus \{0\}$.

At this point, let $I \subseteq \mathbb{R}$ interval, $s \in I \mapsto \gamma(s) \in \mathrm{C}^1(\mathbb{M}^4)$ is called **causal curve (or null-like, time-like, space-like)** if its tangent vectors $\dot{\gamma}(s)$ are causal (or null-like, time-like, space-like).

Definition 3. If we assign a **pseudo orthonormal basis** (e_0, e_1, e_2, e_3), such that $g(e_\mu, e_\nu) = g_{\mu\nu}$, and $[g_{\alpha\beta}] = \eta$, we can see $\mathcal{V}_p$ as disjoint union of two **nappes**:

$$\mathcal{V}_p = \{v^\alpha e_\alpha | (v^0)^2 > \sum_{a=1}^{3} (v^a)^2, v^0 > 0\} \bigsqcup \{v^\alpha e_\alpha | (v^0)^2 > \sum_{a=1}^{3} (v^a)^2, v^0 < 0\} \quad (2)$$

We say that two causal vectors u,v have the same **temporal orientation** if they belong to the topological closure of the same nappe of the lightcone. It can be shown that it is the case if and only if $g(u,v) < 0$.

Definition 4. We assign a temporal orientation to $\mathbb{M}^4$ itself, by associating one of the two nappes of $\mathcal{V}_p$ for every $p \in \mathbb{M}^4$, in a continuous manner, as follows:

- we can assign a vector field X on $\mathbb{M}^4$, which has to be C^0, and such that $g(X_p, X_p) < 0$, and the **temporal orientation** $\mathcal{V}_p^+$ is, at given p, the nappe that contains X_p, and is called **future lightcone**. Causal vectors of $\mathcal{V}_p^+$ are **directed towards the future**. A causal curve $\gamma : I \to \mathbb{M}^4$ is directed towards the future if $\forall s \in I, \dot{\gamma}(s) \in \overline{\mathcal{V}_p^+}$

At this point we can define **proper time**, a parameter proportional to the arc length of a causal curve:

$$\tau(s) = \frac{1}{c} \int_{s_0}^{s} \sqrt{-g(\dot{\gamma}(\xi), \dot{\gamma}(\xi))} d\xi \quad (3)$$

where $\frac{d\tau}{ds} = \frac{1}{c}\sqrt{-g(\dot{\gamma}(s), \dot{\gamma}(s))} > 0$, therefore we can use τ to parametrize the curve. The physical meaning of τ is that it corresponds to the time measured by an ideal clock at rest with the material point moving along γ.

Observation 1. Given two events p,q, linked by two different curves in space-time, synchronization of clocks isn't preserved, simply because the length of the curves isn't necessarily the same.

The tangent vector $V(\tau) = \dot{\gamma}(\tau)$ is called the **four-velocity**.

Minkowskian reference frames and systems of coordinates

Definition 5. A system of coordinates $\psi : p \in \mathbb{M}^4 \mapsto (x^0, x^1, x^2, x^3) \in \mathbb{R}^4$ is called **Minkowskian** if:

- the basis e_0, e_1, e_2, e_3 is pseudo orthonormal
- the timelike vector e_0 is directed towards the future

Definition 6. A **Minkowskian reference frame** is a vector $R \in V^4$, timelike and directed towards the future, such that $g(R, R) = -1$. A Minkowskian system of coordinates ψ is *adapted* to R if $e_0 = R$.

Once R and ψ are fixed, we can "split" spacetime $\mathbb{M}^4$ in time and space:

- we define $t = x^0/c$ as a global time coordinate in R (changing coordinates will modify t by an additive constant);
- (x^1, x^2, x^3) describe the three-dimensional euclidean space at rest with R E_R(changing coordinates will modify (x^1, x^2, x^3) by a rotation and a translation);

Thanks to the previous definition, we can parametrize γ as a function of t. Moreover, we can define a three-velocity, as a function of t, in the space of translations referred to $E_{R,t}$ at a time t: $v^a_{\gamma|R} = \frac{dx^a}{dt}$, a=1,2,3. Since $\tau(t) = \frac{1}{c}\int_{t_0}^{t} \sqrt{-g(\dot{\gamma}(\xi), \dot{\gamma}(\xi))}d\xi$, we can write $V(t)$:

$$V^0(t) = \frac{c}{\sqrt{1 - v(t)^2/c^2}} \quad V^a(t) = \frac{v^a(t)}{\sqrt{1 - v(t)^2/c^2}} \quad v^2 = \sum_{a=1}^{3} (v^a)^2 \tag{4}$$

Now we can easily see that the condition on $V(t)$ to be timelike implies that $v(t) < c$, therefore *the formalism implies* that in any Minkowskian reference frame, the speed must not exceed c. In fact the condition on $\dot{\gamma}(t)$ to be null-like implies that $v^2 = c^2$ in every Minkowskian reference frame. The converse is also true, which means that if the speed of a particle is c in a given reference R, it is null-like, and its speed is c in any other. We have just derived (SR1).

Conservation of the four-momentum We assume that if the worldlines of N material points pass through the same event $p \in \mathbb{M}^4$, originating M new worldlines, there exist positive constants $m_1, ...m_N$ and $m'_1, ...m'_M$ such that:

$$\sum_{j=1}^{N} m_j V_j = \sum_{i=1}^{M} m'_i V'_i \tag{5}$$

where $V_1, ...V_N$ and $V'_1, ...V'_N$ are the four-velocities right before and right after the "scattering".

Definition 7. Given a material, the constant m described by equation 5 is its **mass**, and given its four-velocity parametrized by proper time $V(\tau)$, we can define the **four-momentum** $P(\tau) = mV(\tau)$

Observation 2. Clearly, $g(P,P) = -m^2c^2$, and the fact that m>0 guarantees that the worldline is timelike. Expressing P in coordinates, we find that $P^0 = \sqrt{\sum_{a=1}^{3}(P^a)^2 + m^2c^2}$. In particular, if in some reference frame $P^a = 0 \quad a = 1,2,3$, $P^0 = mc$ carries the information of the mass of the particle.

Let's write equation (5) in a particular system of Minkowskian coordinates adapted to a Minkowskian reference frame:

$$\sum_i \frac{m_i v_i^a}{\sqrt{1 - v_i^2/c^2}} = \sum_j \frac{m_i' {v'}_j^a}{\sqrt{1 - {v'}_j^2/c^2}} \tag{6}$$

If we look at the approximation for $v_i, v_j' \ll c$, at zeroth order we get the classical law of conservation of (three-)momentum. We now consider an example where we keep the first (non-zero) orders in $\frac{v}{c} \ll 1$. We start with a particle of mass m, at rest in some Minkowskian reference frame R. If that particle "splits" into two particles of mass m_1 and m_2, writing equation (6) in R, we get:

$$mc = \frac{m_1 c}{\sqrt{1 - v_1^2/c^2}} + \frac{m_2 c}{\sqrt{1 - v_2^2/c^2}} \quad 0 = \frac{m_1 v_1^a}{\sqrt{1 - v_1^2/c^2}} + \frac{m_2 v_2^a}{\sqrt{1 - v_2^2/c^2}} \tag{7}$$

$$m = \frac{m_1}{\sqrt{1 - v_1^2/c^2}} + \frac{m_2}{\sqrt{1 - v_2^2/c^2}} = m_1 + m_2 + \frac{m_1 v_1^2}{2c^2} + \frac{m_2 v_2^2}{2c^2} + \mathcal{O}((\frac{v_1}{c})^4) + \mathcal{O}((\frac{v_2}{c})^4) \tag{8}$$

Neglecting higher order terms, we can see that equation (8) reduces to:

$$m = m_1 + m_2 + \frac{\mathcal{T}_1}{c^2} + \frac{\mathcal{T}_2}{c^2} \tag{9}$$

We can see that the mass is no longer a conserved quantity in special relativity, and we find that energy and mass are related. In this case if we take the total energy to be conserved, it must be true that $\Delta E = \Delta mc^2$, which means that mass is a form of energy. Moreover, the conservation of the four-momentum $P(\tau)$ implies the conservation of energy, as $cP^0 = \frac{mc^2}{\sqrt{1-(\frac{v}{c})^2}}$ represents the total (mechanical) energy in the reference R.

Relativistic dynamics [2]

Definition 8. A **four-force** is a function $f : (p,v) \in \mathcal{V}^+\mathbb{M}^4 \mapsto f(p,v) \in T_p\mathbb{M}^4$, with the additional condition that $\forall (p,v) \in \Omega \subset \mathcal{V}_p^+\mathbb{M}^4$, with Ω being an open set in which $g(v,v) = -c^2$, f satisfies $g(f(p,v),v) = 0$.

[2]We will consider just the cases where m>0, and take four-forces to be in $\mathcal{V}^+\mathbb{M}^4 = \{(p,v) \in T\mathbb{M}^4 | (p,v) \in \mathcal{V}_p^+\}$

Observation 3. The additional condition requested in the definition 8 is motivated by the following argument. We assume that the relativistic equation of motion is satisfied:

$$\frac{dP}{d\tau} = F(\gamma(\tau), V(\tau)) \tag{10}$$

Additionally we know that we can write $P = mV \implies g(V, P) = -mc^2$, therefore deriving both sides, we get[3]:

$$\frac{dg(P,V)}{d\tau} = -\frac{dP^0}{d\tau}V^0 - P^0\frac{dV^0}{d\tau} + \frac{dP^a}{d\tau}V^a + \frac{dV^a}{d\tau}P^a = 0 = 2g(F,V) \tag{11}$$

This result holds on any solution (p,v) of equation (10). Since we expect that a solution can pass by every (q,u) in the neighboring points of $(p, v) \in \mathcal{V}_p^+\mathbb{M}^4$, with u being timelike and directed towards the future, and also $g(u, u) = -c^2$, we require that there exist $\Omega \subset \mathcal{V}_p^+\mathbb{M}^4$ such that $g(F(q, u), u) = 0 \quad \forall (q, u) \in \Omega$[4].

Theorem 1. Consider a point mass object, of mass m>0 fixed, with timelike worldline $\gamma(s) \in C^1(\mathbb{M}^4)$, which satisfies:

$$m\frac{d\dot{\gamma}}{ds} = F(\gamma(s), \dot{\gamma}(s)) \tag{12}$$

where $F : \mathcal{V}^+\mathbb{M}^4 \to T\mathbb{M}^4$, $F \in C^1(T\mathbb{M}^4)$ is a four-force. Then, for every initial condition $\gamma(s_0) = p$, $\dot{\gamma}(s_0) = v$, such that v is timelike and $g(u, u) = -c^2$, there exist exactly one maximal solution.

Observation 4. If we consider an *isolated* point, for which F=0, P is constant:

$$\gamma(\tau) = \gamma(\tau_0) + (\tau - \tau_0)V(\tau) \implies \gamma(\tau(t)) = \gamma(t_0) + (t - t_0)\sqrt{1 - \left(\frac{v}{c}\right)^2}V(\tau_0) \tag{13}$$

And v is constant in every Minkowskian reference frame, which means that *Minkowskian references frames are inertial references frames.*

Observation 5. We need to stress two important issues. Firstly, if we want to extend the dynamics (in an intrinsic way) to N bodies, we would "need" N proper times! Secondly, if we want to take into account fields, as well as point particles, we need a procedure to include them in the theory so as to make them consistent within the framework of special relativity.

Lorentz and Poincaré groups

In this section we characterize the transformations from one Minkowskian system of coordinates to another, and see the properties of the mappings.

[3] We are assuming m>0 to be constant

[4] The previous property is helpful to demonstrate the following theorem

Definition 9. Given two Minkowskian system of coordinates $\psi : p \in \mathbb{M}^4 \mapsto (x^0, x^1, x^2, x^3) \in \mathbb{R}^4$ and $\psi' : p \in \mathbb{M}^4 \mapsto (x'^0, x'^1, x'^2, x'^3) \in \mathbb{R}^4$, the metrics expressed in the two basis gives:

$$g_{\mu\nu} = g(e_\mu, e_\nu) = \Lambda^\alpha_\mu \Lambda^\beta_\nu g(e'_\alpha, e'_\beta) = \Lambda^\alpha_\mu \Lambda^\beta_\nu g_{\alpha\beta} \tag{14}$$

where $e_\mu = \Lambda^\alpha_\mu e'_\alpha$, and:

$$0 > g(e_0, e'_0) = g(\Lambda^\alpha_0 e'_\alpha, e'_0) = \Lambda^0_0 g_{00} = -\Lambda^0_0 \tag{15}$$

and since the metric g is independent of the basis:

$$\Lambda^T \eta \Lambda = \eta, \quad \eta = diag(-1, 1, 1, 1) \tag{16}$$

Then $\psi \ \circ \psi'^{-1}$ is represented by:

$$x^\alpha = c^\alpha + \Lambda^\alpha_\mu x'^\mu, \quad c^\alpha \in \mathbb{R}, \quad \alpha = 0, 1, 2, 3. \tag{17}$$

The set of matrices that satisfy equation (16) is called **Lorentz group**, or $\mathcal{O}(1,3)$, and is a group with respect to composition. If we consider also the translations $(c^0, c^1, c^2, c^3) \in \mathbb{R}^4$, the group of transformation described by equation (17) is called **Poincaré group**, or $\mathcal{IO}(1,3)$. If the condition $\Lambda^0_0 > 0$ is satisfied, they are also **orthochronous**: $\mathcal{O}(1,3)_+$ and $\mathcal{IO}(1,3)_+$.

Definition 10. $\mathcal{IO}(1,3)_+$ includes **internal transformations** subgroups such as spacetime translations $x^\alpha = c^\alpha + x'^\alpha$ and space rotations:

$$\Omega_R = \left(\begin{array}{c|ccc} 1 & 0 & 0 & 0 \\ \hline 0 & & & \\ 0 & & R & \\ 0 & & & \end{array}\right)$$

where $R \in \mathcal{O}(3)$.

Definition 11. $\mathcal{O}(1,3)_+$ contains **Lorentz special transformations**, along directions e_1, e_2, e_3:

$$\Lambda_1(v) = \begin{pmatrix} \gamma & \gamma v/c & 0 & 0 \\ \gamma v/c & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \Lambda_2(v) = \begin{pmatrix} \gamma & 0 & \gamma v/c & 0 \\ 0 & 1 & 0 & 0 \\ \gamma v/c & 0 & \gamma & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\Lambda_3(v) = \begin{pmatrix} \gamma & 0 & 0 & \gamma v/c \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \gamma v/c & 0 & 0 & \gamma \end{pmatrix}, \quad v \in (-c, c), \gamma = \frac{1}{\sqrt{1 - (\frac{v}{c})^2}}$$

These transformations show how coordinates change between two Minkowskian reference frames moving at relative speed v (along e_1, e_2 or e_3).

Observation 6. Although the subgroup 11 is not additive in v, which means that $\Lambda_k(v)\Lambda_k(v') \neq \Lambda_k(v+v')$, we can introduce the **rapidity** χ, defined such that $v(\chi) = c\tanh\chi$. For example, $\Lambda_1(\chi)$ is:

$$\Lambda_1(v(\chi)) = \begin{pmatrix} \cosh\chi & 0 & 0 & \sinh\chi \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh\chi & 0 & 0 & \cosh\chi \end{pmatrix}$$

Written in this form, it is true that $\Lambda_k(v(\chi))\Lambda_k(v(\chi')) = \Lambda_k(v(\chi+\chi'))$.

Observation 7. Thanks to Lorentz special transformations 11, it can be seen easily that the concept of contemporaneity is dependent on the Minkowskian reference frame. Moreover, if two events in spacetime are spacelike, different inertial observers can see their time order inverted. We can conclude that a *total order relation*[5] of all events doesn't exist.

Theorem 2. Given $\Lambda \in \mathcal{O}(1,3)_+$, there exist $R_1, R_2 \in \mathcal{O}(3)$ and $v \in (-c,c)$ such that:[6]

$$\Lambda = \Omega_{R_1}\Lambda_3(v)\Omega_{R_2} \tag{18}$$

where Ω_{R_1} and Ω_{R_2} are defined as in definition 10.

Observation 8. Thanks to theorem 2, we can limit ourselves to the study of Lorentz special transformations, as any transformation in $\mathcal{IO}(1,3)_+$ is a composition of internal transformations and a Lorentz special transformation.

Lagrangian formalism

We will briefly discuss some Lagrangian approaches to describe point particles in the presence of an (external) EM field in special relativity, and present their strength and weaknesses. Introducing the **four-EM-potential** (in a Minkowskian system of coordinates):

$$A(p) = A^\alpha e_\alpha = \varphi(p)e_0 + A^a(p)e_a \in T_p\mathbb{M}^4, p \in \mathbb{M}^4 \tag{19}$$

which is covariant: changing coordinates it transforms into $A'^\mu = \Lambda^\mu_\nu A^\nu$.

Quadratic covariant Lagrangian

$$\mathcal{L}_1(\tau,\gamma,\dot\gamma) = \frac{m}{2}g(\dot\gamma,\dot\gamma) + \frac{e}{c}g(\dot\gamma, A(\gamma)) \tag{20}$$

Observation 9. This way of constructing the Lagrangian is purely geometrical, it doesn't depend on the coordinates. We can write $\mathcal{L}_1$ in a given system of

[5] thought as independent of the reference frame

[6] the same theorem holds for Lorentz special transformations along e_1, e_2 as well.

coordinates:

$$\begin{aligned}\mathcal{L}_1(s,x,\dot{x}) &= \frac{m}{2}g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu + \frac{e}{c}g_{\mu\nu}\dot{x}^\mu A^\nu(x) \\ &= \frac{m}{2}\left(-(\dot{x}^0)^2 + \sum_{a=1}^{3}(\dot{x}^a)^2\right) - \frac{e}{c}\dot{x}^0\varphi(x) - \frac{e}{c}\sum_{a=1}^{3}\dot{x}^a A^a(x)\end{aligned} \tag{21}$$

if we take s to be proper time and consider $v \ll c, \quad \dot{x}^0 \approx c$, we get the classical Lagrangian:

$$\mathcal{L}_1 \approx \frac{m}{2}\sum_{a=1}^{3}\left(\frac{dx^a}{dt}\right)^2 - e\varphi(x(t)) + ec\sum_{a=1}^{3}\frac{dx^a}{dt}A^a(x(t)) - \frac{mc^2}{2} \tag{22}$$

Solving Euler-Lagrange equations, we get:

$$\sum_{\alpha=0}^{3} g_{\alpha\beta}m\frac{d^2x^\alpha}{ds^2} = \sum_{\alpha=0}^{3}\frac{e}{c}\frac{dx^\mu}{ds}F_{\mu\beta}(x(s)), \quad F_{\mu\nu} = \frac{\partial A_\mu}{\partial x^\nu} - \frac{\partial A_\nu}{\partial x^\mu} \tag{23}$$

where $A_\mu = g_{\mu\nu}A^\nu$. Since $[g_{\mu\nu}]$ is nonsingular, equation (23) can be written in normal form, thus we can have a theorem of uniqueness of solutions thanks to Cauchy theorem.

Observation 10. We can now focus on symmetries and conserved quantities, considering just the kinetic part: $\mathcal{L}_1(\tau,\gamma,\dot{\gamma}) = \frac{m}{2}g(\dot{\gamma},\dot{\gamma})$. Since $\frac{\partial \mathcal{L}_1}{\partial s} = 0$, applying Jacobi's theorem, the quantity $\mathcal{H}(s,\gamma,\dot{\gamma}) = \frac{m}{2}g(\dot{\gamma},\dot{\gamma})$ is conserved. $\mathcal{L}_1$ is invariant under these one-parameter groups:

- temporal translations[7]: $x'^0_\epsilon = x^0 + \epsilon$
- spatial translations along e_1[8]: $x'^1_\epsilon = x^1 + \epsilon$
- spatial rotations around $e_3$the same transformation can be done around e_1 and e_2: $x'^1_\epsilon = x^1\cos\epsilon - x^2\sin\epsilon, \quad x'^2_\epsilon = x^1\sin\epsilon + x\cos\epsilon$
- Lorentz special transformations along e_3[9]:

  $x'^0_\epsilon = x^0\cosh\epsilon + x^3\sinh\epsilon, \quad x'^3_\epsilon = x^0\sinh\epsilon + x^3\cosh\epsilon$

It can be seen easily that $\mathcal{L}_1$ is invariant under these one-parameter groups, therefore we have the following conserved quantities, thanks to Noether theorem:

$$I(s,x,\dot{x}) = \sum_{\mu=0}^{3}\frac{\partial \mathcal{L}_1}{\partial \dot{x}^\mu}\frac{\partial x'^\mu_\epsilon}{\partial \epsilon}\bigg|_{\epsilon=0} \tag{24}$$

We note that Lorentz special transformations give a new conserved quantity, called **boost vector K**, such that: $K^a = x^0P^a - x^aP^0$, where P is the momentum.

[7] we show here only coordinates that change when subject to the one-parameter groups
[8] the same transformation can be done along e_2 and e_3
[9] the same transformation can be done along e_1 and e_2

Non-quadratic covariant Lagrangian

$$\mathcal{L}_2(s,\gamma,\dot\gamma) = -mc\sqrt{-g(\dot\gamma,\dot\gamma)} + \frac{e}{c} g(\dot\gamma, A(\gamma)) \tag{25}$$

Observation 11. The Lagrangian is defined if $g(\dot\gamma,\dot\gamma) < 0$, which means that γ is directed towards the future. It can be easily seen that if $v \ll c$, the Lagrangian reduces to its non relativistic analog. Writing down Euler-Lagrange equations:

$$mc\sum_{\alpha=0}^{3} g_{\alpha\beta}\frac{d}{ds}\left(\frac{1}{\sqrt{-\sum_{\mu,\nu} g_{\mu\nu}\frac{dx^\mu}{ds}\frac{dx^\nu}{ds}}}\frac{dx^\alpha}{ds}\right) = \sum_{\nu=0}^{3}\frac{e}{c}\frac{dx^\nu}{ds}F_{\nu\beta}\Big|_{x(s)} \tag{26}$$

The equations, as formulated in equation (26), don't satisfy the condition for a theorem of uniqueness of solutions to apply. Anyway, it can be shown that every solution belongs to the same curve in $\mathbb{M}^4$. Deriving $\mathcal{L}_2$ with respect to s gives $\frac{\partial \mathcal{L}_2}{\partial s} = 0$, which means that Jacobi theorem holds, however $\mathcal{H} = 0$. As a result, a Hamiltonian can't be constructed starting from $\mathcal{L}_2$.

Non-quadratic non-covariant Lagrangian Considering a Minkowskian system of coordinates and time t, we can write the following Lagrangian:

$$\mathcal{L}_3 = -mc^2\sqrt{1-\sum_{a=1}^{3}\left(\frac{\dot x^a}{c}\right)^2} - \frac{e}{c}\varphi(x) + \frac{e}{c}\sum_{a=1}^{3}\dot x^a A^a(x) \tag{27}$$

Observation 12. Similarly to observation 11, the Lagrangian is defined if $1 - \sum_{a=1}^{3}\left(\frac{\dot x^a}{c}\right)^2 > 0$, and for $v \ll c$ the classical Lagrangian is recovered. Euler-Lagrange equations are:

$$m\frac{d}{dt}\left(\frac{1}{\sqrt{1-\sum_b \frac{1}{c^2}(\frac{dx^b}{dt})^2}}\frac{dx^a}{dt}\right) = -e\frac{\partial\varphi}{\partial x^a} + \frac{e}{c}\sum_{a=1}^{3}\frac{dx^b}{dt}F_{ab} \tag{28}$$

The equations, along with initial conditions directed towards the future, satisfy a theorem of uniqueness of solutions.

Observation 13. In the case of e=0, the *action functional* doesn't depend on coordinates, unlike $\mathcal{L}_3$:

$$I[\gamma] = -mc^2\int_{t_1}^{t_2}\sqrt{1-\sum_{a=1}^{3}\frac{1}{c^2}\left(\frac{dx^a}{dt}\right)^2}\,dt \tag{29}$$

$$= -mc\int_{t_1}^{t_2}\sqrt{\left(\frac{dx^0}{dt}\right)^2 - \sum_{a=1}^{3}\left(\frac{dx^a}{dt}\right)^2}\,dt \tag{30}$$

$$= -mc\int_{s_1}^{s_2}\sqrt{-g(\dot\gamma(s),\dot\gamma(s))}\,ds \tag{31}$$

Observation 14. If φ and A don't depend explicitly on time, we have $\frac{\partial \mathcal{L}_3}{\partial t} = 0$, and the function $\mathcal{H} = \frac{mc^2}{\sqrt{1-\sum_{a=1}^{3}\left(\frac{\dot{x}^a}{c}\right)^2}} + e\varphi(t, x^1, x^2, x^3)$ is constant. In absence of electromagnetic fields, $\mathcal{H} = cP^0$

Observation 15. Since the formulation of $\mathcal{L}_3$ requires the global time t and not proper time (like in $\mathcal{L}_1$ and $\mathcal{L}_2$, the formalism can be extended to N point particles. We can write the Lagrangian as the sum of the Lagrangians of every point particle:

$$\mathcal{L}_3 = \sum_{i=1}^{N}\left(- m_i c^2 \sqrt{1 - \sum_{a=1}^{3}\left(\frac{\dot{x}_i^a}{c}\right)} - e_i\varphi(t, x_i) + \frac{e_i}{c}\sum_{a=1}^{3}\dot{x}^a A^a(t, x_i)\right) \tag{32}$$

We now focus on the kinetic part, neglecting external fields. Integrals of motion can be written as the sum of the integral of motion of every point particle:

$$P^a = \sum_{i=1}^{N}\frac{m_i\dot{x}_i^a}{\sqrt{1-\sum_{b=1}^{3}\left(\frac{\dot{x}_i^b}{c}\right)^2}}, \quad \mathcal{H} = \sum_{i=1}^{N}\frac{m_i c^2}{\sqrt{1-\sum_{b=1}^{3}\left(\frac{\dot{x}_i^b}{c}\right)^2}} = \sum_{i=1}^{N}\mathcal{H}_i \tag{33}$$

Which implies, thanks to $\frac{d\mathcal{H}_\rangle}{dt} = 0$:

$$P^a\Bigg|_{E-L} = \frac{d}{dt}\Bigg|_{E-L}\sum_{i=1}^{N}\frac{\mathcal{H}_i x_i^a}{c^2} \tag{34}$$

We can define a *relativistic center of mass*:

$$x_G^a = \frac{1}{\mathcal{H}}\sum_{i=1}^{N}\mathcal{H}_i x_i^a \implies P^a\Bigg|_{E-L} = \frac{d}{dt}\Bigg|_{E-L}\frac{\mathcal{H}x_G^a}{c^2} \tag{35}$$

which means that N point (non-interacting) particles can be treated (when dealing with the momentum) as a single point particle of mass $\mathcal{H}/c^2$ and position x_G.

References

[1] Valter Moretti, *Meccanica Analitica*, Springer, 2024.